

# Nebula@ BITS Goa

25 SEPT | 9 AM

### WHO CAN PARTICIPATE?

Maximum team size: 3  
(students only)

### EVENT HIGHLIGHTS

Tech Talks • Fun Games  
Networking • Career Coaching

## Topics

- 1. AI/ML for Analog Circuit Design
- 2. Constraint Optimization through RTL Enhancement Using Generative AI
- 3. CXL-Based Memory Optimization for MoE Models

## Important Dates

|                       |           |
|-----------------------|-----------|
| Submit Abstract       | 6th Aug   |
| Abstract Shortlisting | 12th Aug  |
| Final Submission      | 15th Sept |
| Final Shortlisting    | 19th Sept |
| Final Presentations   | 25th Sept |

## Prizes

|                          |                   |                          |
|--------------------------|-------------------|--------------------------|
| 1st Runner Up<br>₹30,000 | Winner<br>₹50,000 | 2nd Runner Up<br>₹20,000 |
|--------------------------|-------------------|--------------------------|

“Got the skills? You might just have a shot at the Internship.”

## CALL FOR PARTICIPATION

Semiconductor Industry is where cutting-edge tech meets bold talent -  
Come for the insights, stay for the vibes!

# Constraint Optimization through RTL Enhancement Using Generative AI

---

## Problem Statement

Achieving timing closure is one of the biggest challenges in modern chip design. Engineers spend significant effort analyzing timing reports, identifying critical paths, and manually modifying RTL to meet performance targets - a time-consuming process requiring multiple design iterations. There is an opportunity to leverage Generative AI to analyze timing bottlenecks, recommend RTL optimizations, and accelerate timing closure while preserving functional correctness.

## Objectives

**Develop a GenAI-assisted framework to:**

- Analyze RTL designs against specified timing constraints
- Identify critical paths and timing violations
- Use GenAI to recommend RTL optimizations (pipelining, logic restructuring, retiming, FSM optimization)
- Evaluate timing, area, and performance improvements
- Formally verify equivalence between original and optimized RTL
- Ensure the RTL benchmark includes: 5 independent master asynchronous clock domains; at least 1 generated clock per master; Clock Domain Crossings (CDC); clock-divider logic with multiple ratios; and approximately 50K standard cells

## Deliverables

- RTL timing analysis framework
- GenAI-based RTL optimization engine
- Critical-path and timing-violation analysis
- Optimized RTL implementation
- Timing, frequency, and PPA comparison
- Formal equivalence verification report
- Interactive demo showcasing the RTL optimization workflow

## Tools & Resources

- Verilog/SystemVerilog
- OpenSTA, Yosys, OpenROAD
- SymbiYosys / EQY
- Python
- LLM / GenAI frameworks